FOODVERSE AI

Multimodal Food Intelligence Agent

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Steps

Activity 1: Requirements Specification and Environment Setup

Activity 1.1: Create Requirements.txt Approach

This activity involved preparing the requirements.txt file to list the Python packages needed by Foodverse AI. Keeping dependencies in one file makes the project environment easier to set up consistently.

Activity 1.2: Install Required Libraries and Setup Virtual Environment

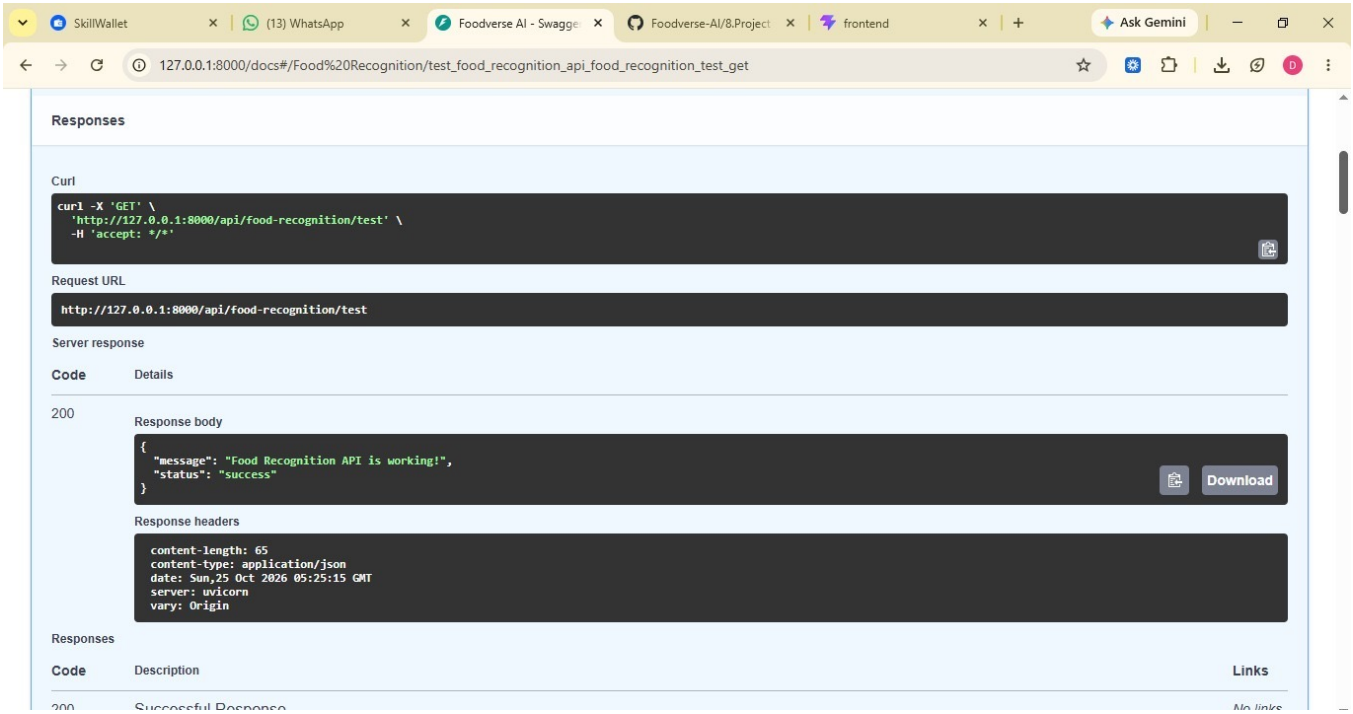
This activity involved creating a Python virtual environment and installing the required libraries. The environment keeps project packages separate and prepares the system for backend development and testing.

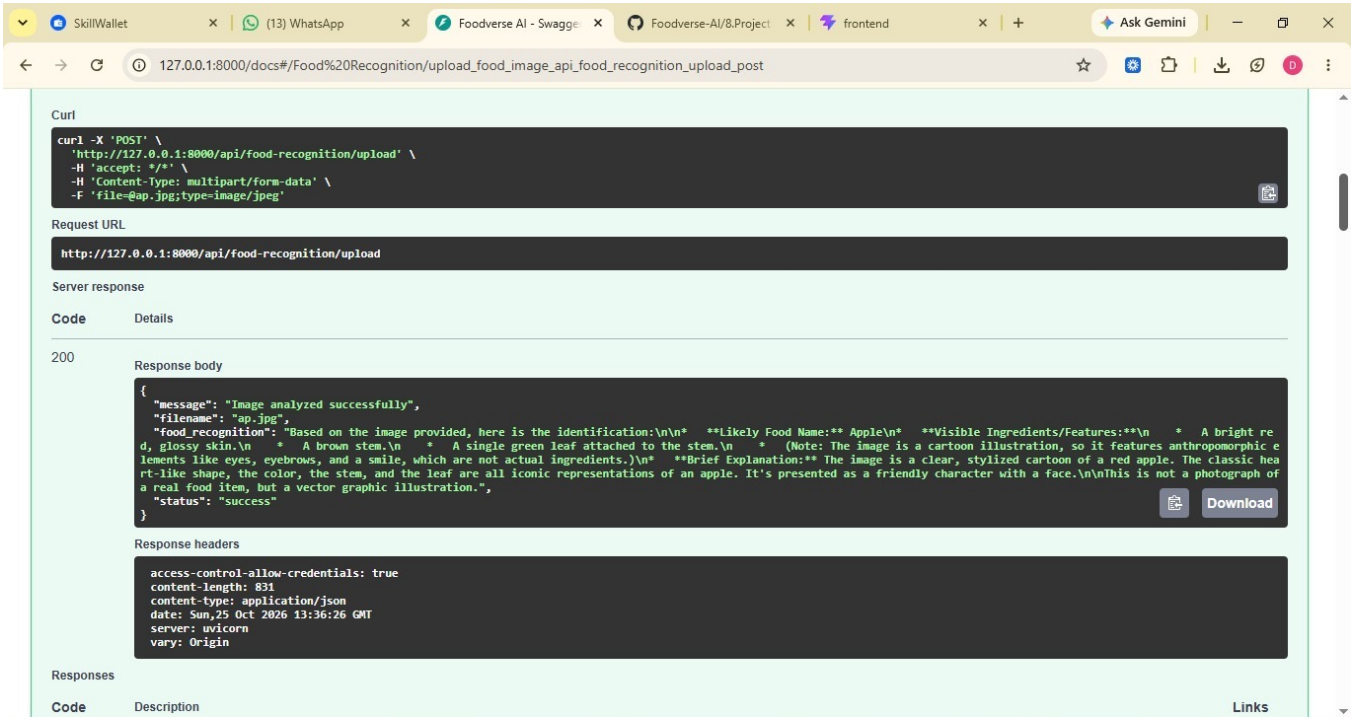
Steps

Activity 2: Food Recognition and Nutrition Module Development

Activity 2.1: Food Image Detection and Classification

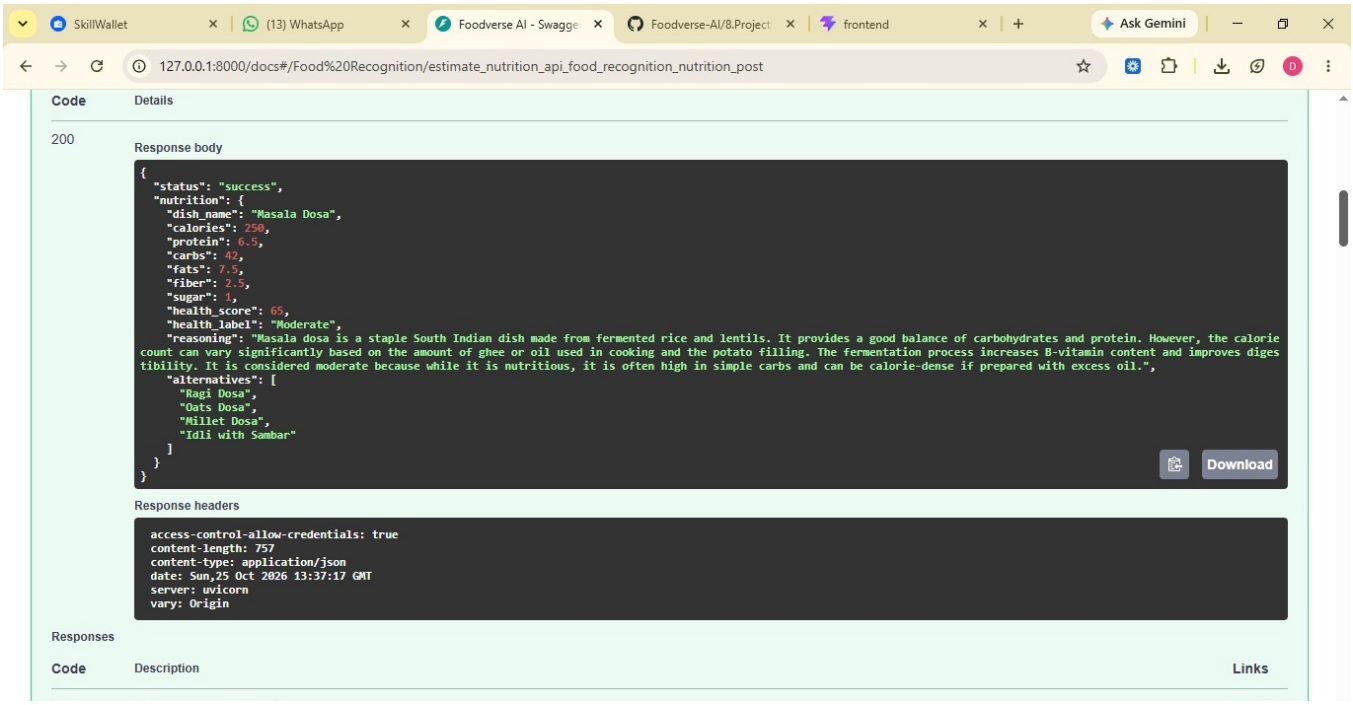
This activity focuses on identifying food items from an uploaded image and classifying the detected food.





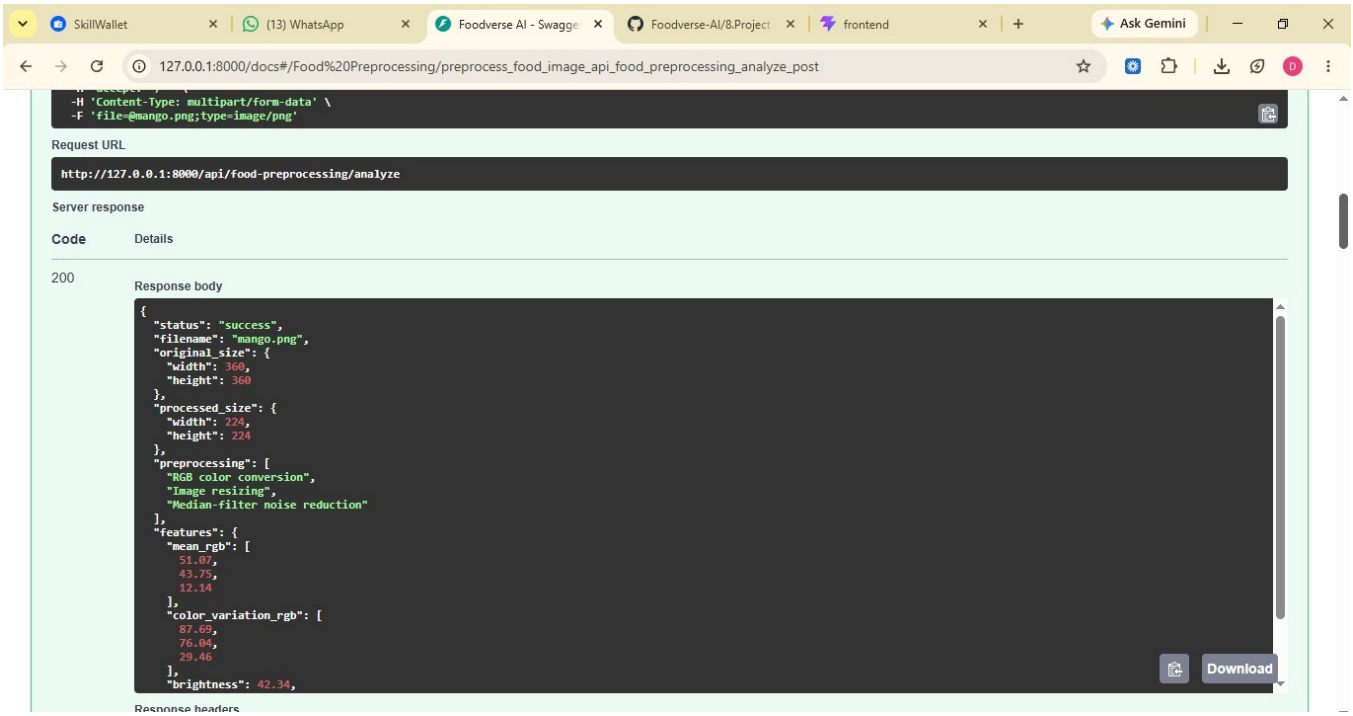
Activity 2.2: Nutrition Estimation and Health Scoring Logic

This activity estimates nutritional information and health-related scoring for recognized food.



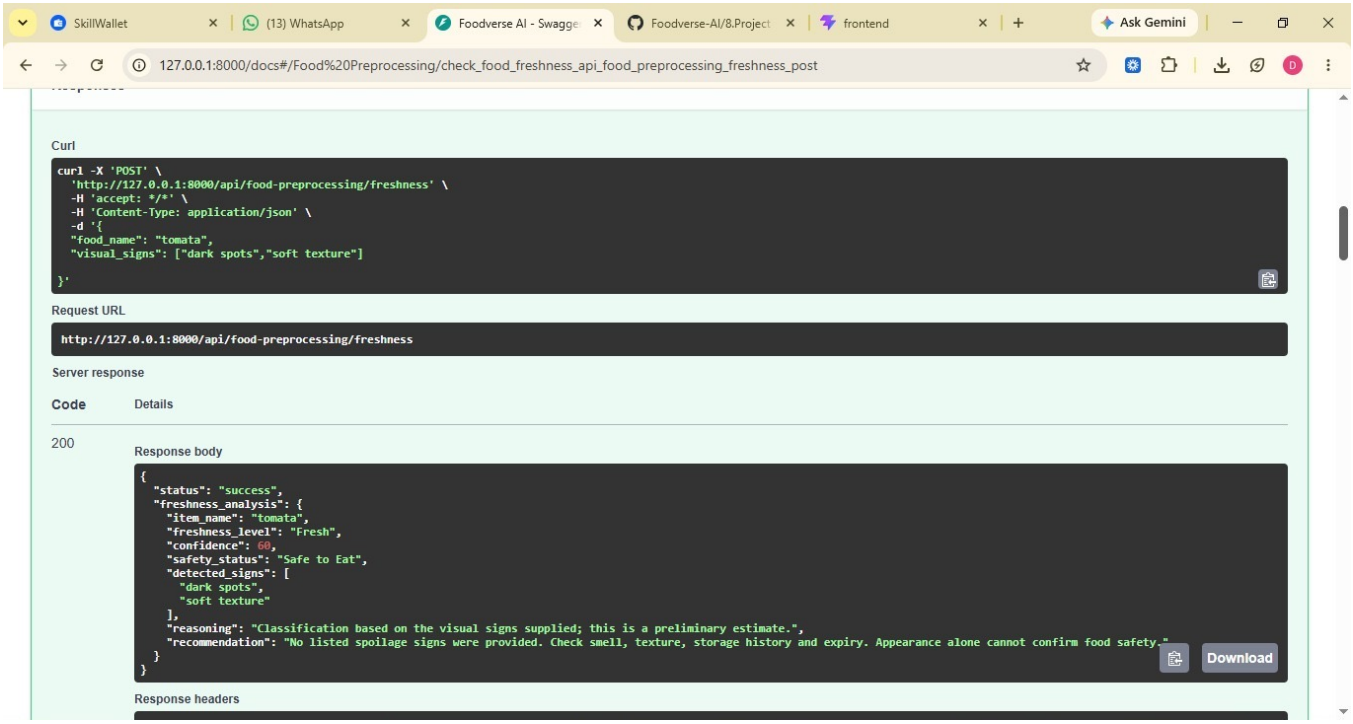
Activity 3: Food Freshness and Quality Detection

Activity 3.1: Food Image Processing and Feature Extraction



This activity prepares food images for analysis by processing the image and extracting useful features.

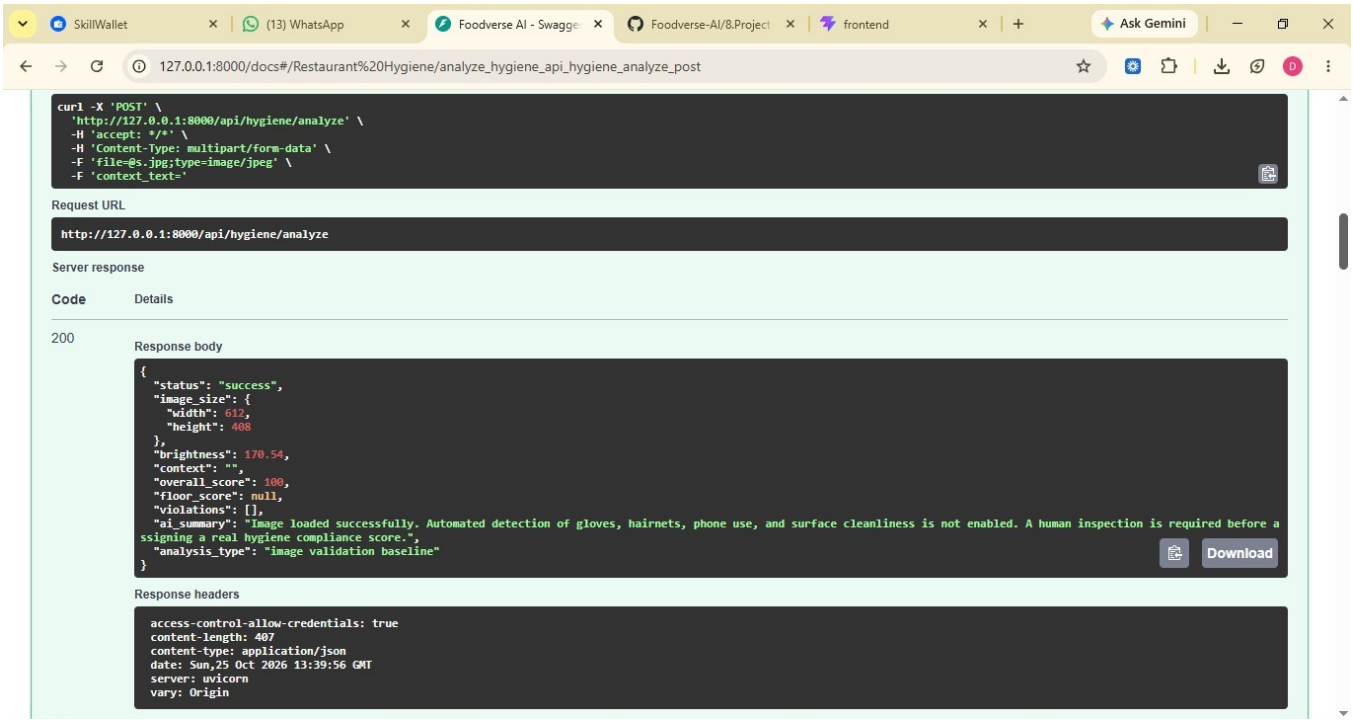
Activity 3.2: Freshness Classification and Safety Recommendation Logic



This activity classifies food freshness and provides a safety recommendation based on the analysis.

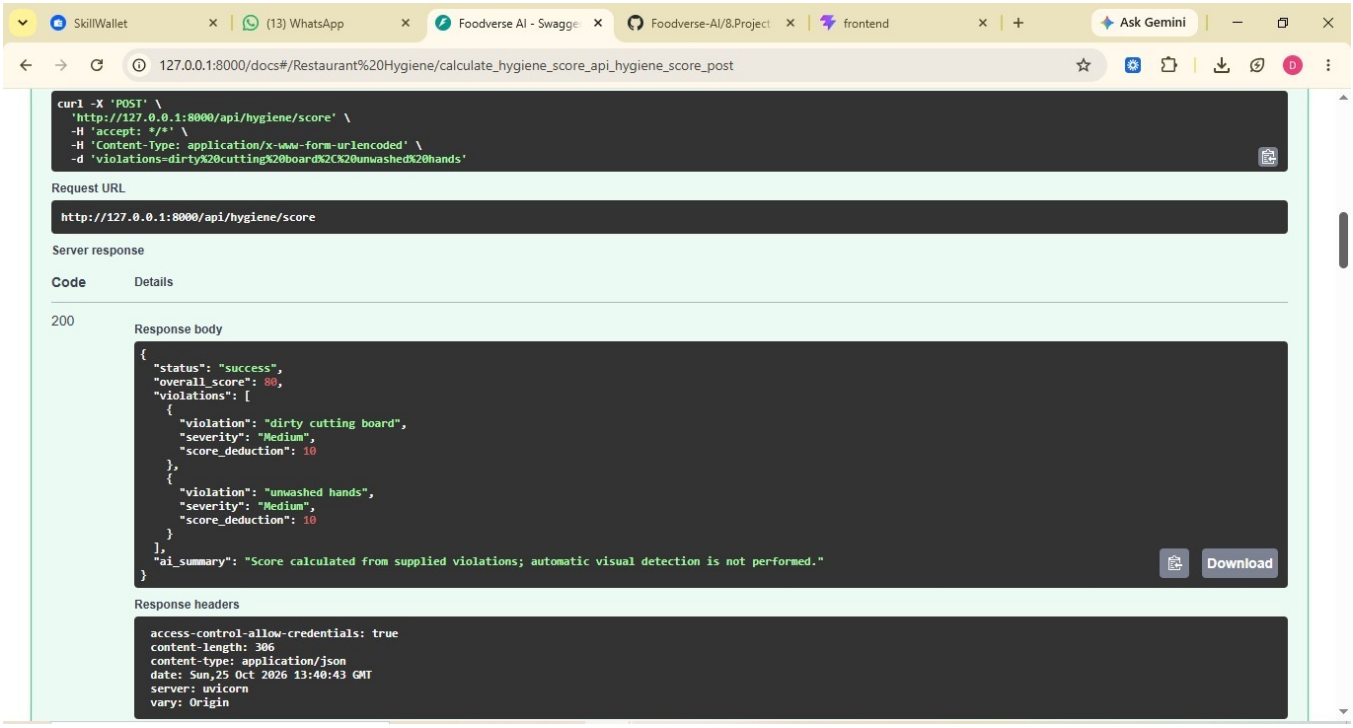
Activity 4: Restaurant Hygiene and Smart Kitchen

Activity 4.1: Restaurant Hygiene Detection and Compliance Scoring



This activity checks restaurant or kitchen hygiene indicators and supports a compliance assessment.

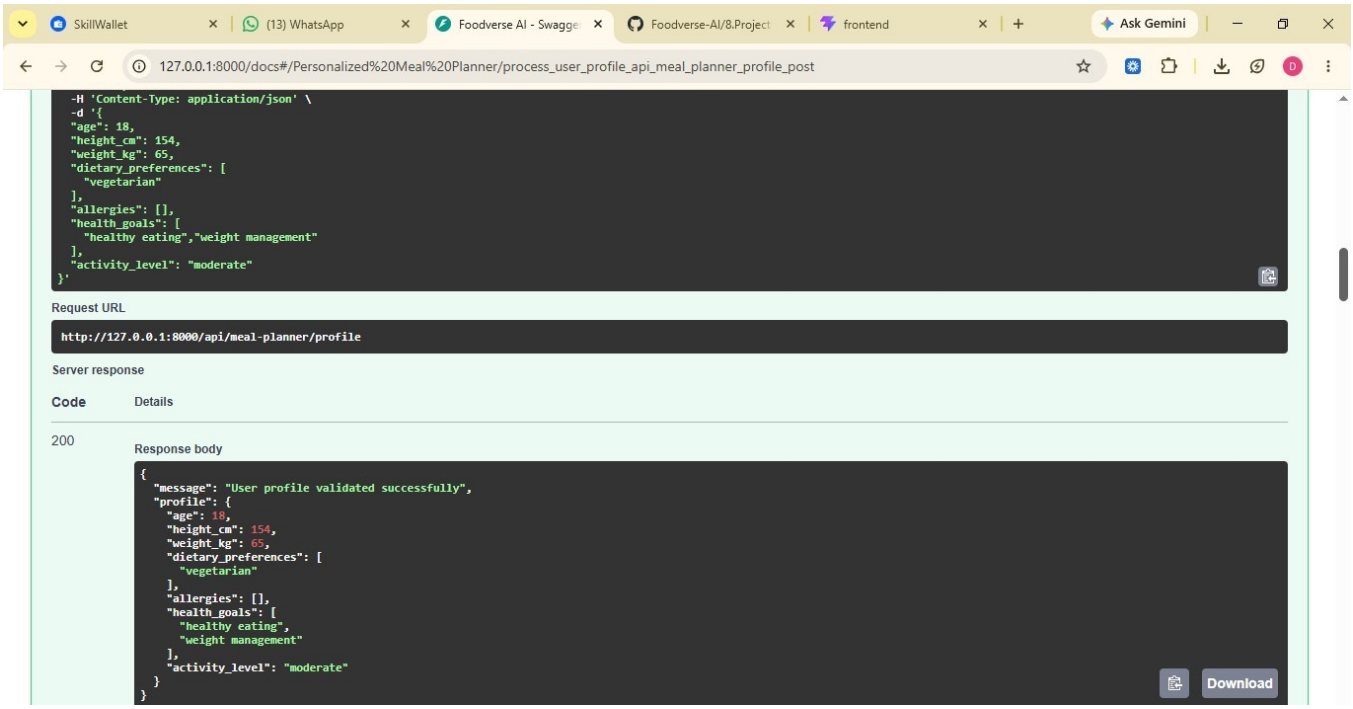
Activity 4.2: Smart Kitchen Cooking Guidance and Recipe Generation Logic



This activity provides smart-kitchen cooking guidance and generates recipes based on the available inputs.

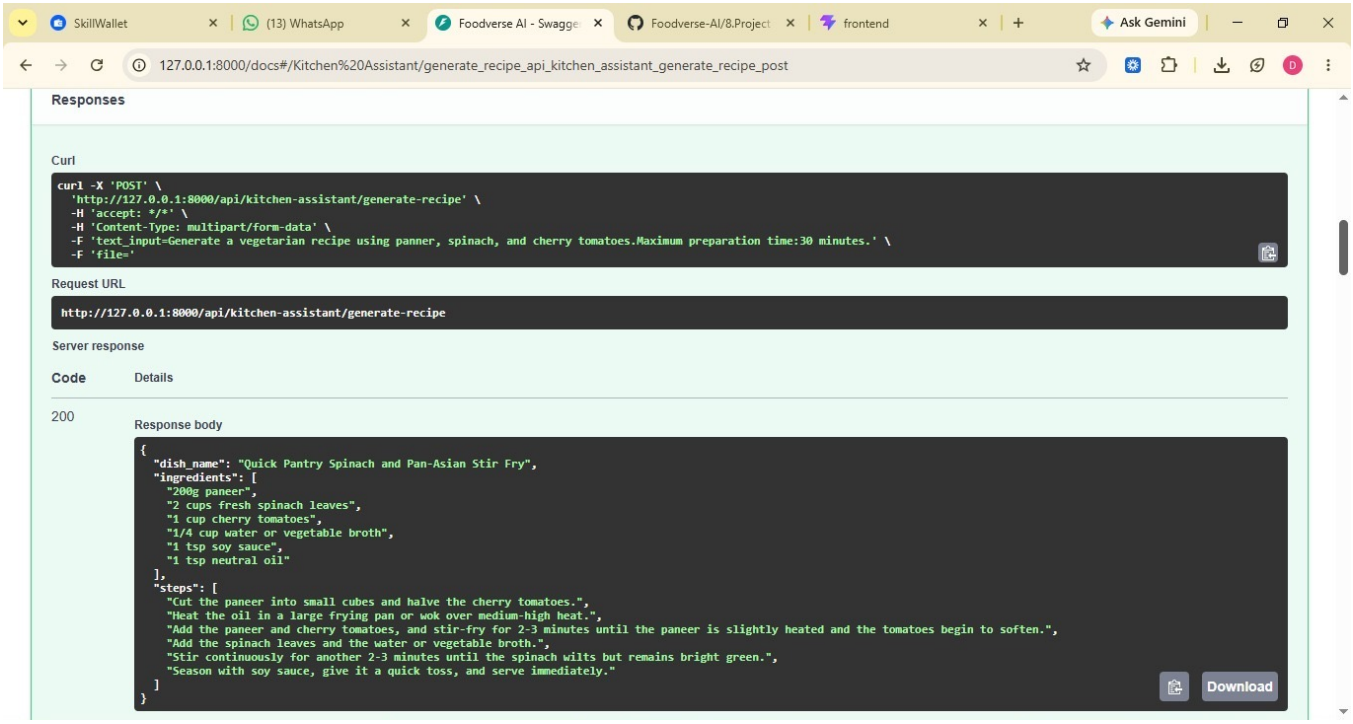
Activity 5: Personalized Meal Plan Development

Activity 5.1: User Profile, Goal and Preference Processing



This activity processes the user's profile, goals, and preferences to support personalized meal planning.

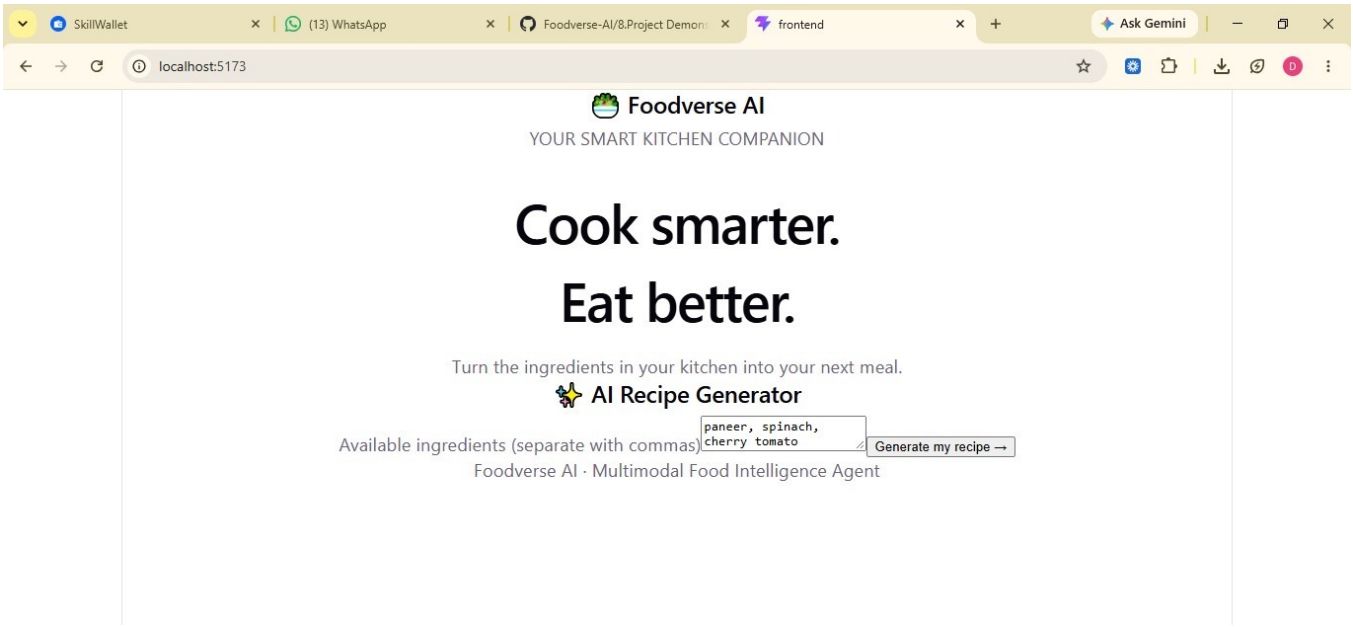
Activity 5.2: Meal Plan Generation and Adaptive Recommendation Logic



This activity generates meal plans and adapts recommendations to the user's preferences and needs.

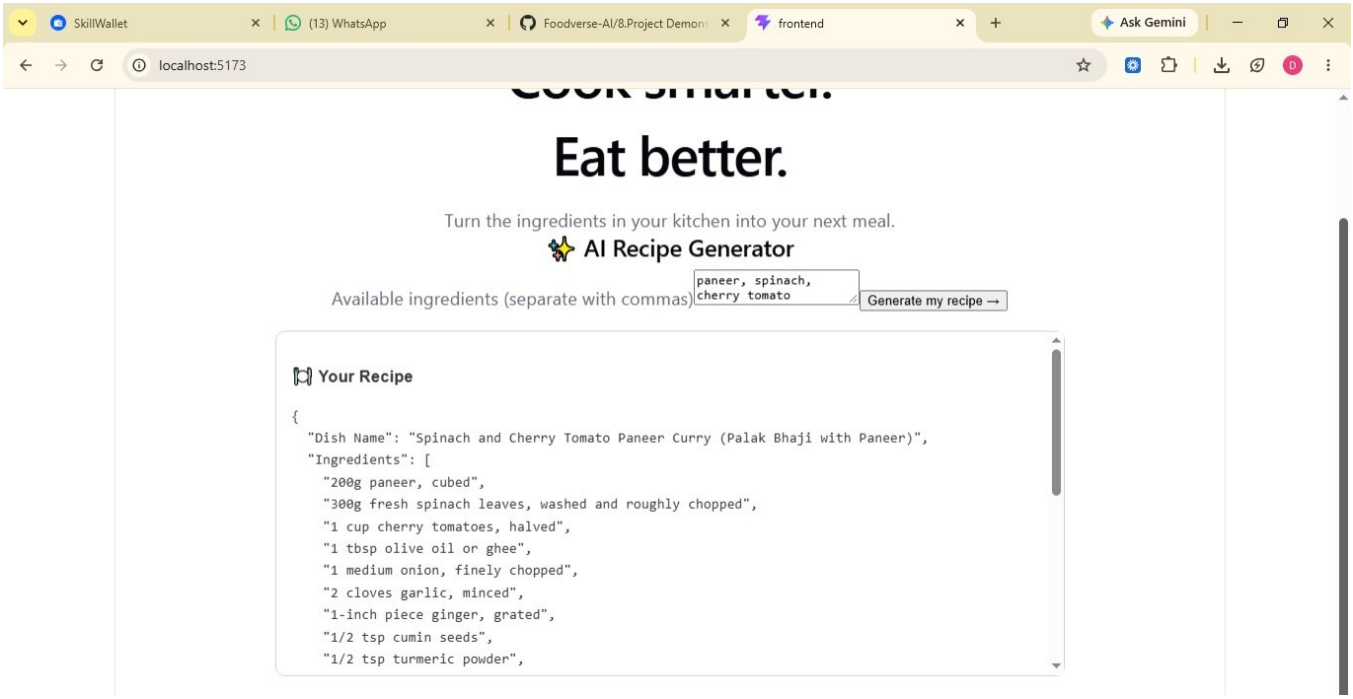
Activity 6: Interference Testing and Deployment

Activity 6.1: Front-End Interface Integration and System Execution



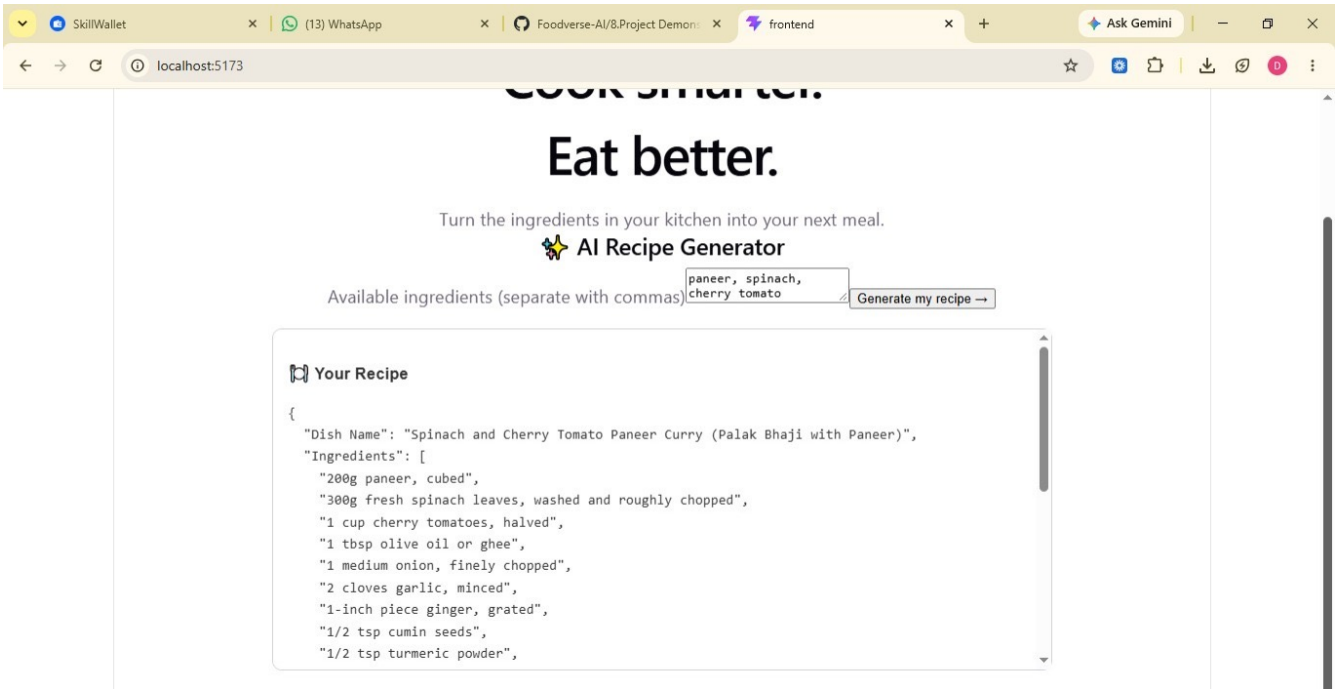
This activity connects the front-end interface with the backend and checks that the integrated system runs.

Activity 6.2: Functional Testing and Edge Case Validation



This activity tests the main functions with different inputs and checks how the system handles edge cases.

Activity 6.3: Final Validation and Deployment Readiness



This activity reviews the integrated application and checks its readiness for deployment.

Conclusion

Foodverse AI was developed as a multimodal food intelligence agent to make cooking smarter and more convenient. The project involved requirement analysis, environment setup, food recognition, nutrition estimation, freshness detection, hygiene monitoring, and personalized meal planning. A backend was developed and tested, and a front-end interface was integrated to allow users to generate recipes from available ingredients. Functional testing was performed to check the system’s responses and functionality. Overall, this project combines AI and web technologies to create a smart kitchen companion.